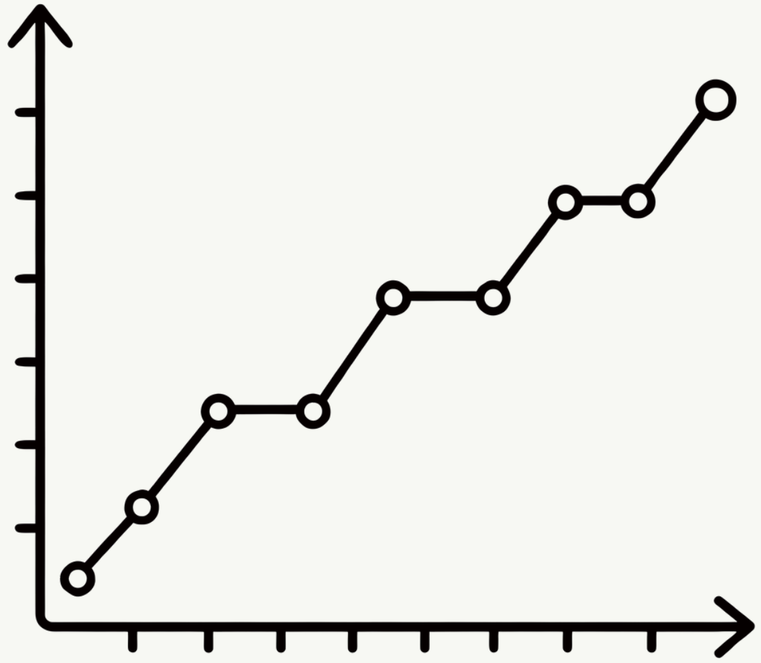
Lime Cost/Quality Scaling on High-Dimensional Tabular Data on Multiple Models

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Overview

1. **Topic & Motivation**

2. **The Core Technical Content:**
- LIME
- Logistic Regression
- XGBoost
- Random Forest
- TabFM

3. **A Critical Angle:**
- Background research
- Experimental Methods
- Runtime Cost
- Stability
- Fidelity

4. **Conclusion**

5. **References**

Topic & Motivation.

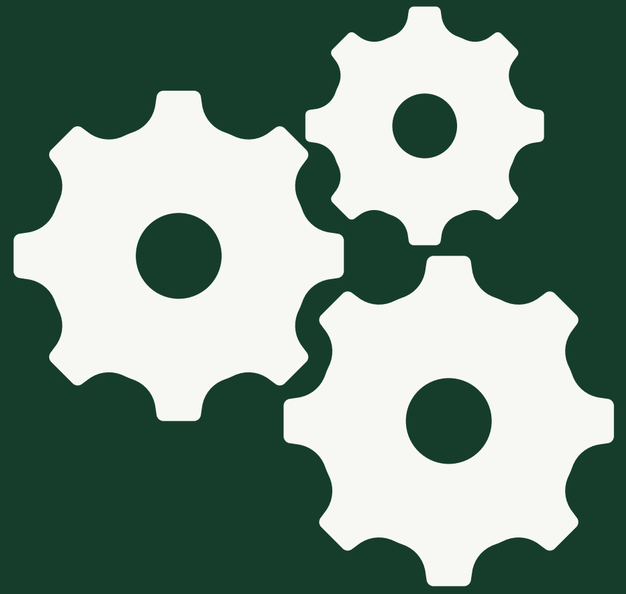
Topic: **Lime Cost/Quality Scaling on High-Dimensional Tabular Data on Multiple Models**

Motivation:

The runtime of LIME depends on the number of perturbation samples and inference cost of the black-box model, while the quality of explanations, especially stability and local fidelity, degrade as feature space grows.

Studies have examined LIME's instability and proposed variations upon LIME, but there is a lack of guidance on how cost and quality might scale together as dimensionality and model complexity increase.

The Core Technical Content

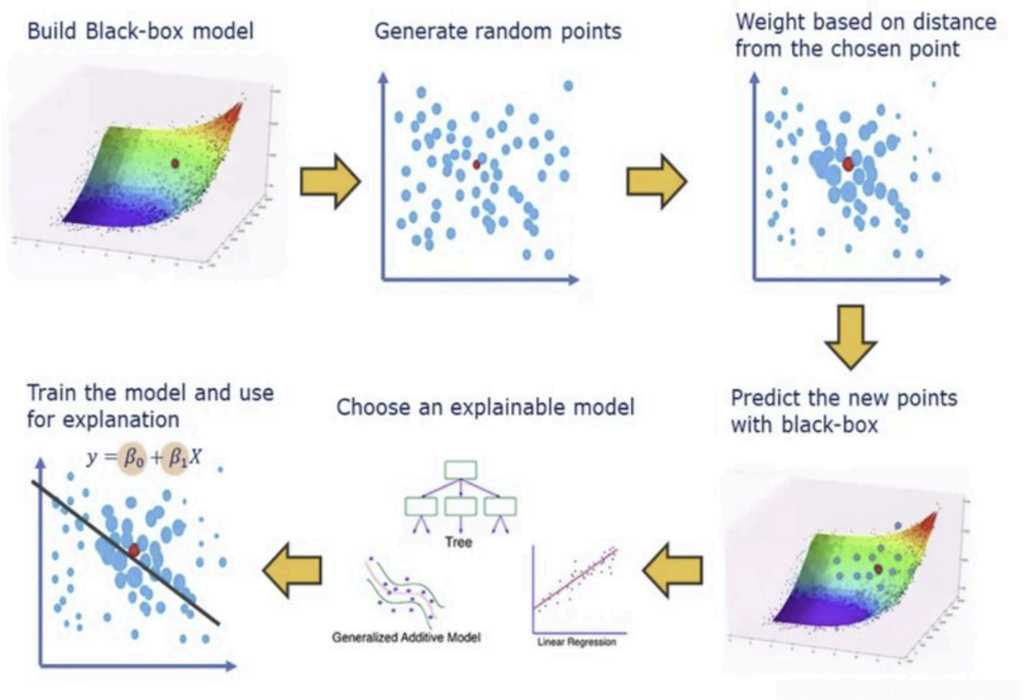


LIME

LIME: Local Interpretable Model-agnostic Explanations is a popular method for explaining predictions made by black box models (GeeksForGeeks 2025).

The algorithm works as follows:

- Choose your ML model and a reference point/prediction to be explained.
- Generate random points.
- Make predictions with your chosen original model
- Give more importance to similar points.
- Train a simple linear model
- LIME fits a simple weighted linear regression model to the generated examples.
- Instead of being complex, this model uses a simple equation.
- LIME looks at the coefficients of the simple model to explain which features increase or decrease the predicted price near the original house (Visani, G 2020).



Logistic Regression

How it works:

- Supervised classification model predicting class probabilities using weighted features.
- Classifies seven forest cover types.

Why we chose it:

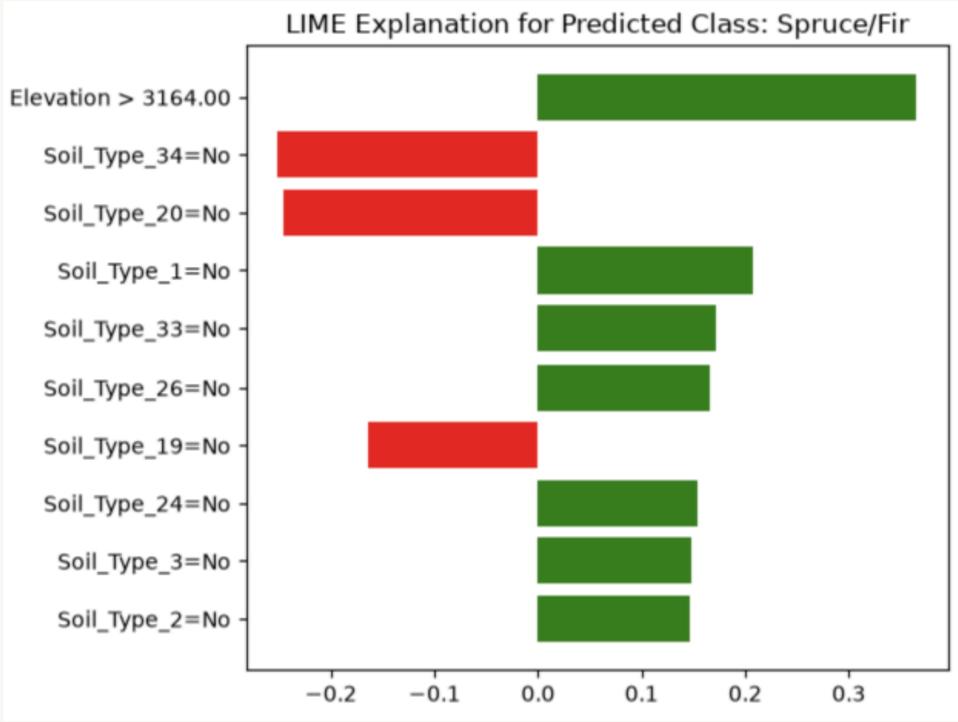
- Simple, efficient and interpretable.
- Provides a baseline for comparing LIME against more complex models.

Pros

- Clear display of features support or opposition
- Allows comparison with local LIME explanations and global model coefficient
- Fast max sample size took 0.047 seconds

Cons

- Computationally expensive due to repeated predictions
- Explians individual predictions but not overall model behaviour
- Explainations can vary depending on sample



Opposes Prediction VS Supports Prediction
Test Accuracy: 72.34%

Random Forest

The Random Forest algorithm is an ensemble learning technique, using multiple decision trees to make predictions.

How it works:
Each tree trains on a random sample of data. The results of each tree are combined by voting for classification or averaging for regression. (GeeksForGeeks 2026; Samy Baladram 2024)

Why we chose it:

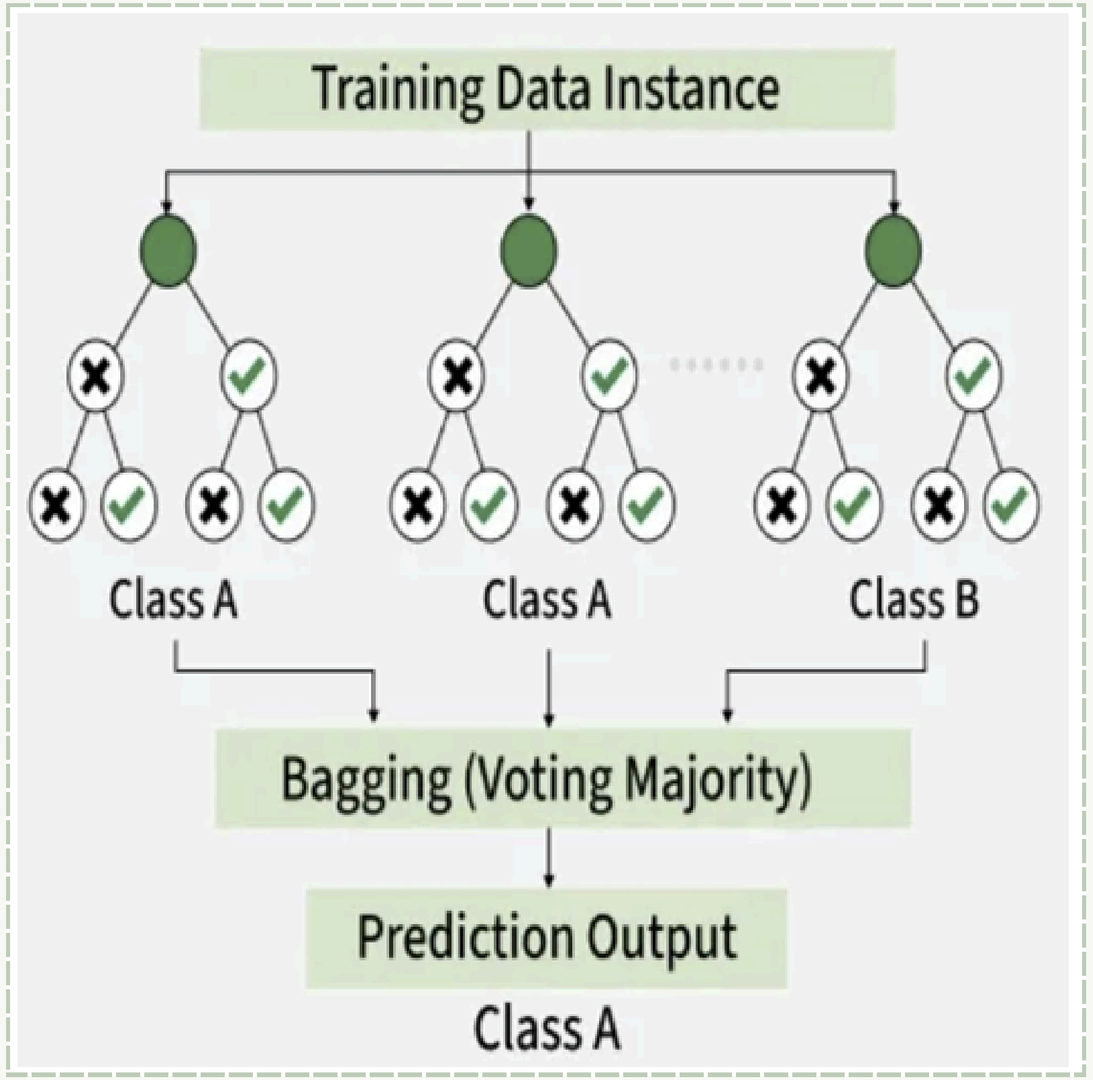
- Contrasts Logistic Regression - a simple linear model - well.
- Tests LIME on a more complex model

Pros:

- Higher accuracy
- Resistant to overfitting

Cons:

- Computational complexity
- Less direct interprability



XGBoost

What is XGBoost?

- A gradient-boosted ensemble of decision trees.
- Captures nonlinear relationships and feature interactions.

How does it work?

- Builds decision trees sequentially.
- Each new tree improves the existing model's predictions.

Why did we choose it?

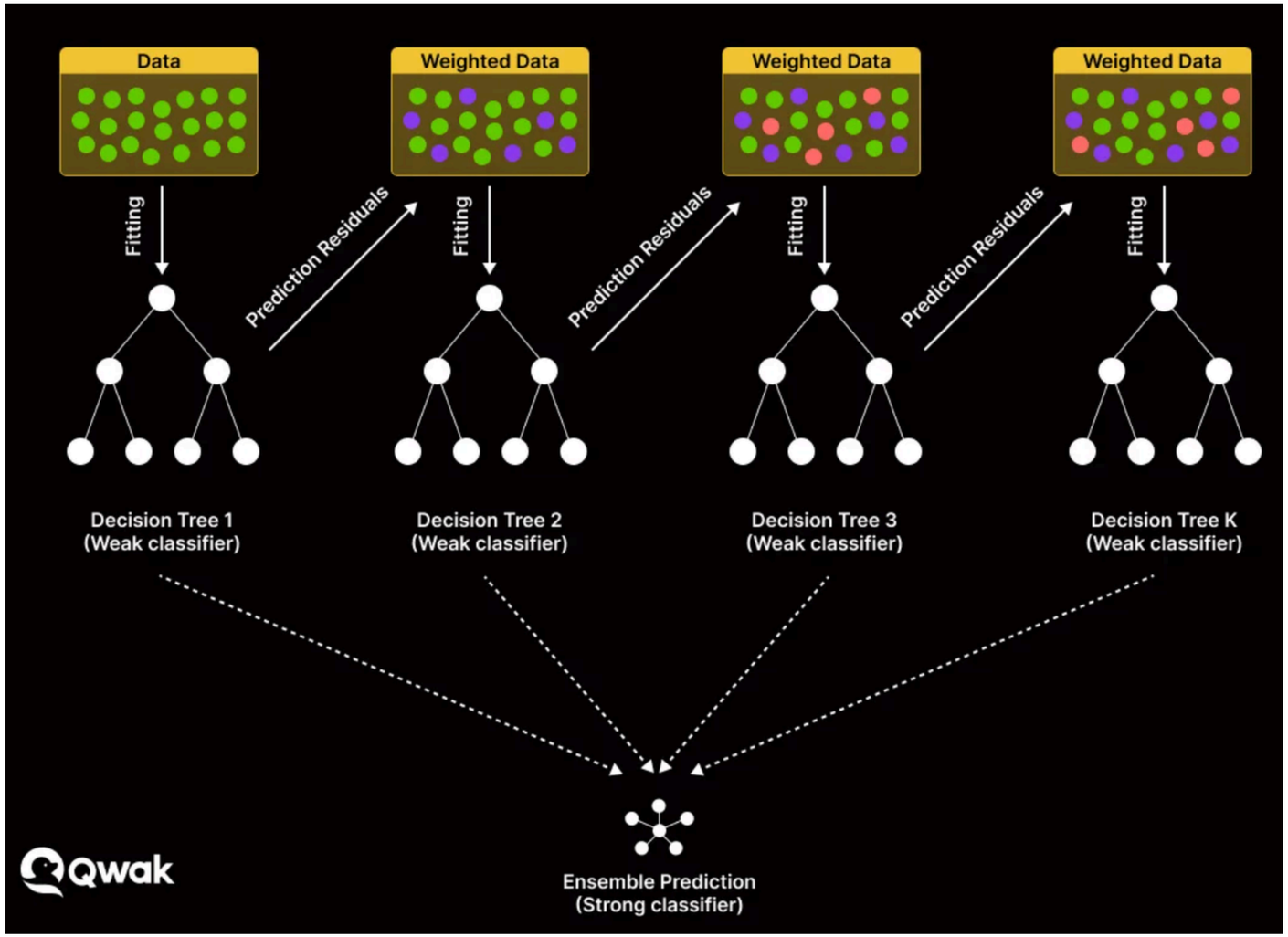
- Strong performance on structured/tabular data.
- Tests LIME on a complex, nonlinear model.

Advantages:

- Effective on structured data.
- Captures complex feature relationships.

Limitations:

- Difficult to interpret directly.
- Training and hyperparameter tuning can be complex.



TabFM

TabFM: Tabular Foundation Model | Created by Google Research

- A zero-shot () foundation model
- Designed for two-dimensional tables of rows and columns (tabular data)
- 400-million-parameter model, trained by Google on millions of synthetic tables

How it works:
TabFM is pre-trained on millions of synthetic tables, so instead of training on our data, it reads our labelled rows as context and predicts new rows from the patterns it spots

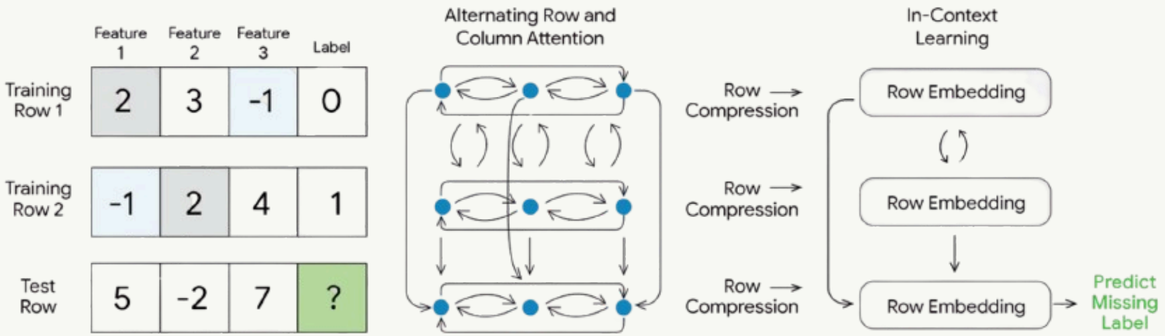
Why we chose it:
LIME explains a prediction by asking the model thousands of slightly different questions. Each of TabFM's answers is expensive, so it's the costliest of our four models to explain, which makes it the best test of LIME's cost versus quality

Pros:

- No training or tuning
- Accurate with little data (thousands rather than hundred of thousands)
- Probabilities you can trust

Cons:

- Expensive to predict with
- Can't use big datasets
- Trained on made-up data



A Critical Angle



Background Research

• Between us 8 papers were covered, and we broke our findings into 3 themes:

Theme 1: LIME is expensive, but nobody has measured that systematically on tabular data.

- *A. S. Ross, M. C. Hughes, and F. Doshi-Velez, "Right for the right reasons: Training differentiable models by constraining their explanations," in Proc. 26th Int. Joint Conf. Artificial Intelligence (IJCAI), 2017, pp. 2662–2670. doi: 10.24963/ijcai.2017/371.*
- C. Mougan Navarro, "Model monitoring in the absence of labeled data via feature attributions distributions," Ph.D. dissertation, Univ. of Southampton, Southampton, U.K., 2025. [Online]. Available: https://eprints.soton.ac.uk/498652/
- K. K. Bladen, A. Cutler, D. R. Cutler, and K. R. Moon, "Conditional local importance by quantile expectations," Transactions on Machine Learning Research, Jun. 2026. [Online]. Available: https://openreview.net/forum?id=gsuZFPDRqE

Theme 2: more samples make LIME more stable, but cost more. Only qualitatively, or for one model.

- X. Zhao, W. Huang, X. Huang, V. Robu, and D. Flynn, "BayLIME: Bayesian local interpretable model-agnostic explanations," in Proc. 37th Conf. Uncertainty in Artificial Intelligence (UAI), PMLR, vol. 161, 2021, pp. 887–896. [Online]. Available: https://proceedings.mlr.press/v161/zhao21a.html
- C. Burger, C. Walter, T. Le, and L. Chen, "Towards robust and accurate stability estimation of local surrogate models in text-based explainable AI," arXiv:2501.02042, 2025. doi: 10.48550/arXiv.2501.02042.
- P. Knab, S. Marton, U. Schlegel, and C. Bartelt, "Which LIME should I trust? Concepts, challenges, and solutions," in Explainable Artificial Intelligence (xAI 2025), CCIS, vol. 2577. Springer, 2026, pp. 28–52. doi: 10.1007/978-3-032-08324-1_2.

Theme 3: tabular LIME has been evaluated, but not for cost.

- J. Dieber and S. Kirrane, "Why model why? Assessing the strengths and limitations of LIME," arXiv:2012.00093, 2020. doi: 10.48550/arXiv.2012.00093.
- M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), 2016, pp. 1135–1144. doi: 10.1145/2939672.2939778.

Metrics we observed to identify how quality degrades as cost scales.

- Runtime
- Stability
- Fidelity

Experimental Methods

Dataset: Covertype

- ~580,000 rows
- 54 input features
- 7 classification types

Fixed Experimental Settings:

- 80/20 Training Split
- Random State: 42
- Lime Version: 0.2.0.1
- Features Explained: 10
- Repetitions: 5 per Sample Size
- Background Data: 50000 observations

Sample Sizes

- 500
- 1000
- 2500
- 5000
- 10000

Evaluation Metrics:

Runtime

- Seconds

Explanation Stability

- Spearman
- Top 5 Agreement

Local Fidelity

- Limes exp.score

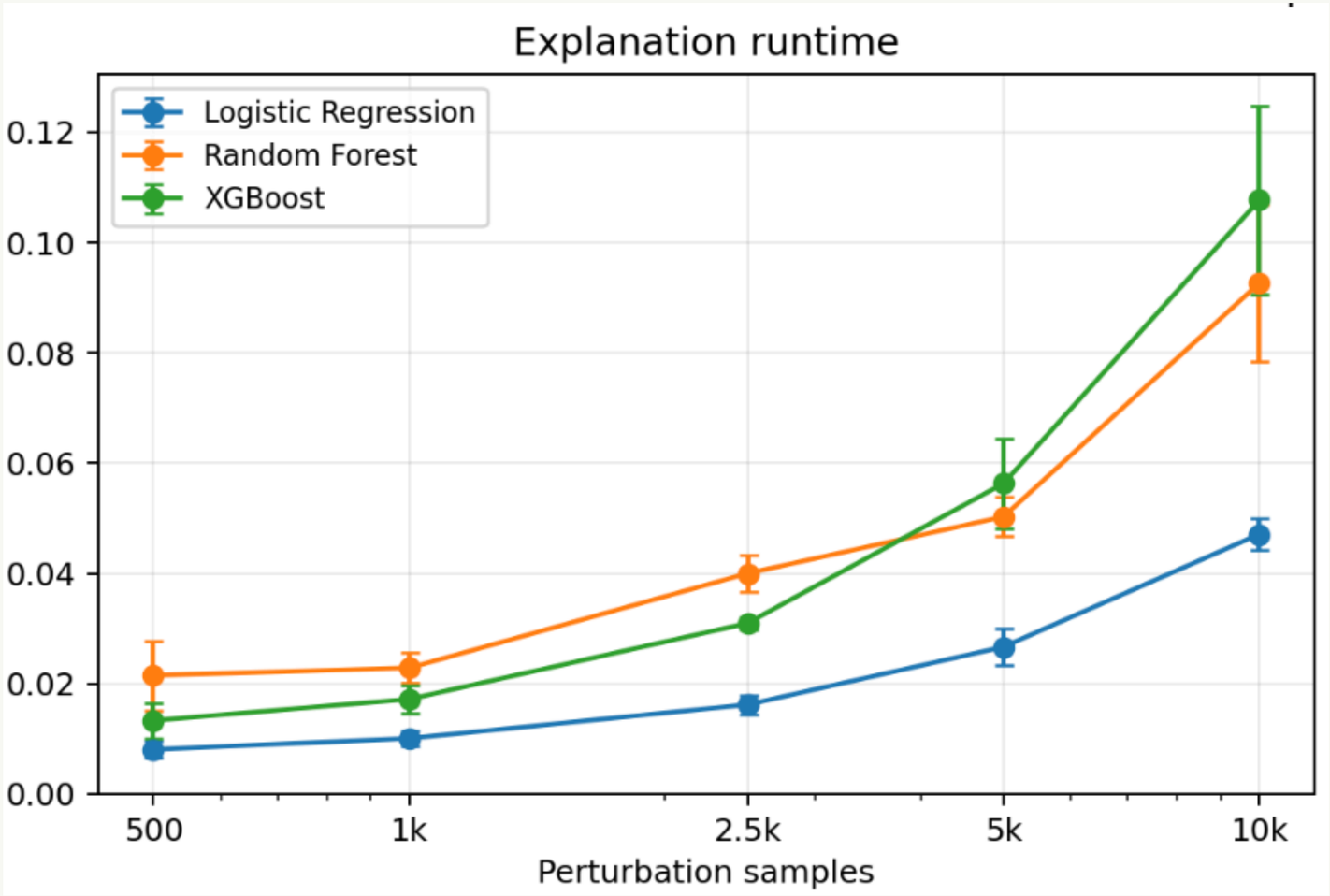
Runtime Cost

Key Finding:
Runtime increases with perturbation samples.

All three models become slower as LIME's sample count increases.

- Logistic Regression: Fastest (~0.047s at 10,000 samples).
- XGBoost: Moderate runtime (~0.108s).
- Random Forest: Moderate runtime (~0.09s).

Takeaway:
Increasing samples introduces additional computational cost, which must be justified by improvements in explanation quality.

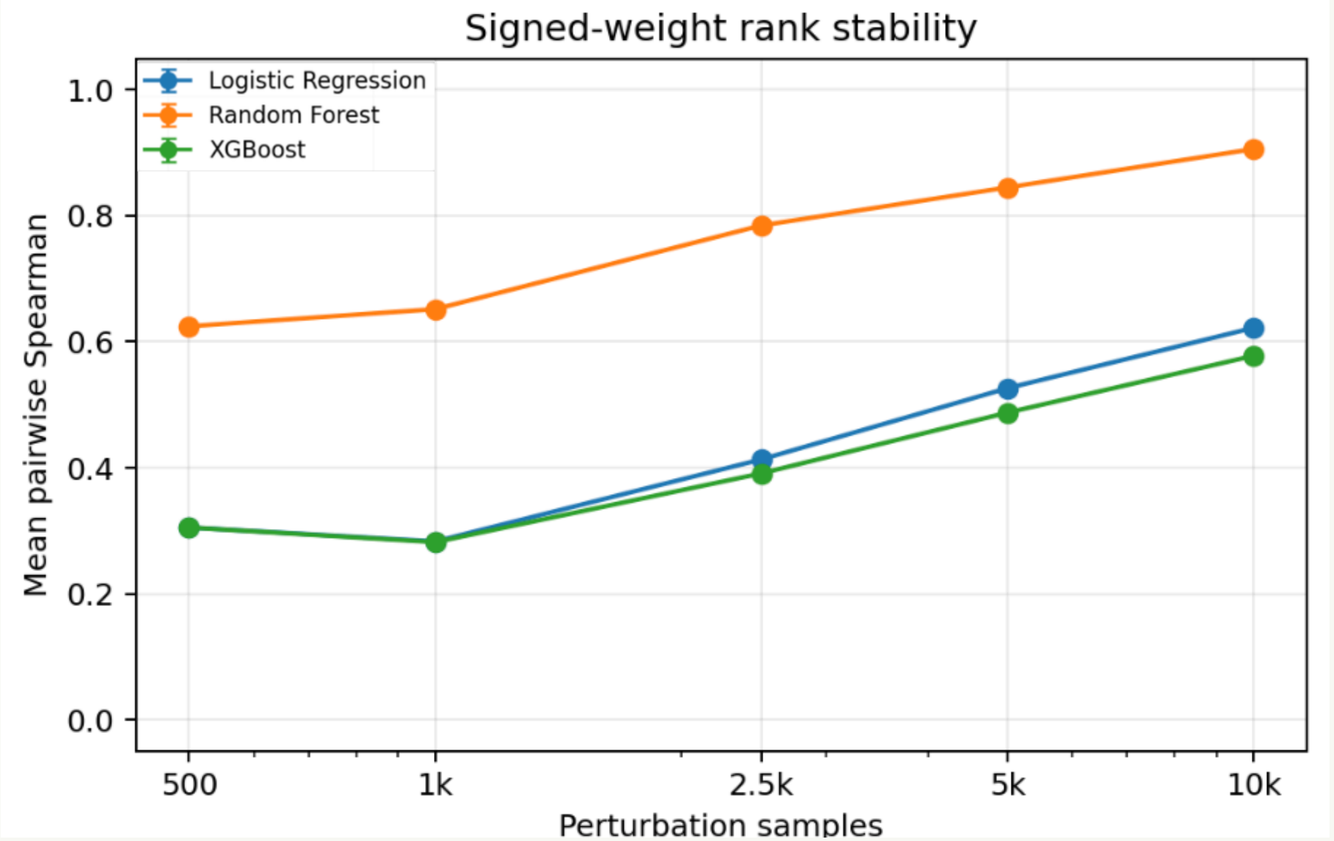


Stability

Key Finding:
More samples means greater stability.

- Random Forest: Highest stability, reaching approximately 0.90 Spearman at 10,000 samples.
- Logistic Regression: Moderate stability, reaching approximately 0.62.
- XGBoost: Lowest stability, reaching approximately 0.58, slightly behind Logistic Regression.

Takeaway:
Random Forest produced the most consistent rankings of signed feature contributions in our experiment, despite having the highest runtime.

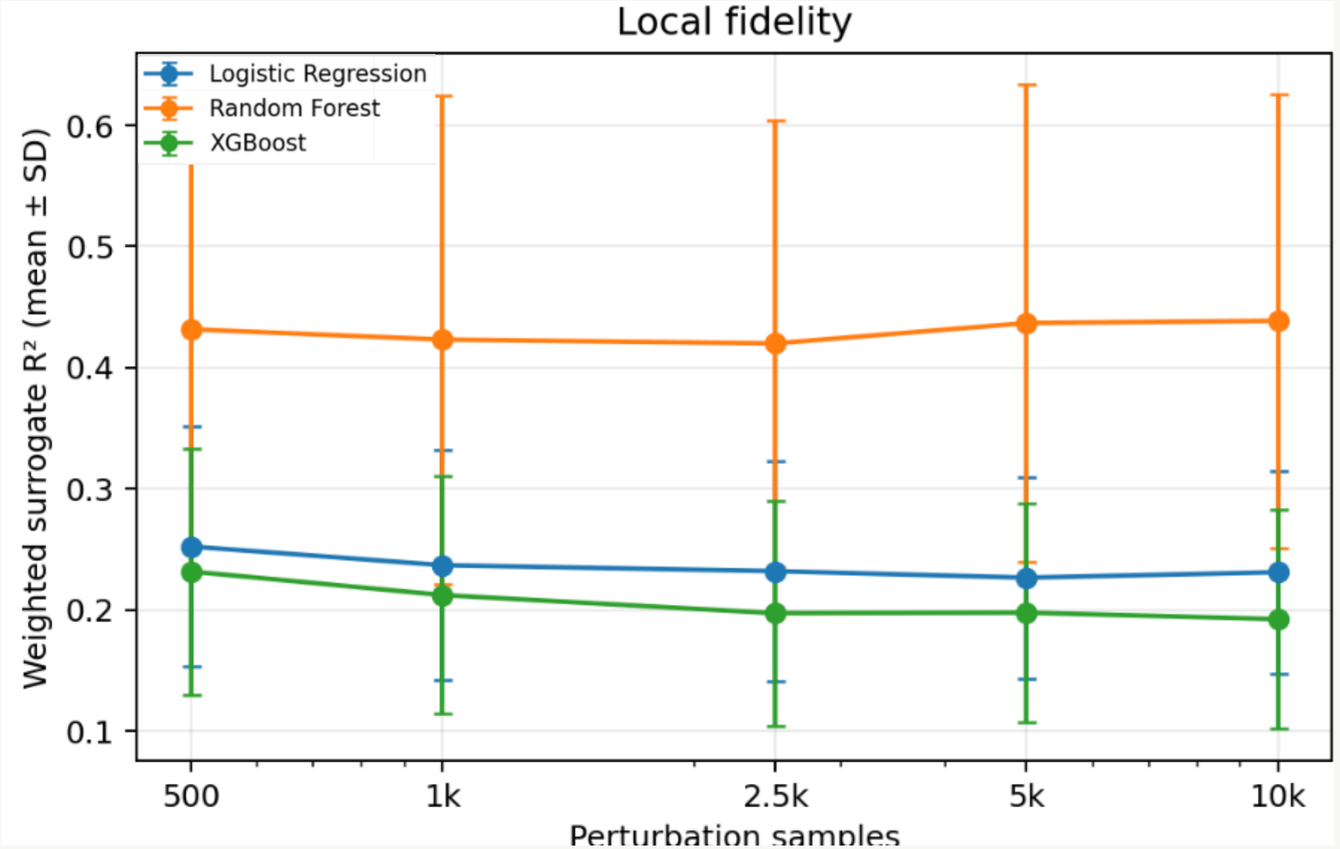


Fidelity

Key Finding:
Increasing perturbation samples produced little-to-none improvement in local fidelity across the three models.

- Random Forest: Highest fidelity, reaching approximately 0.44 weighted R^2 at 10,000 samples.
- Logistic Regression: Moderate fidelity, approximately 0.23
- XGBoost: Lowest fidelity, approximately 0.19 at 10,000 samples.

Takeaway:
Random Forest produced the most faithful local approximations in our experiment, although its results showed greater variability. Increasing sample sizes offered minimal fidelity improvements despite increasing runtime costs.



Conclusion

Key Findings:
Our experiments demonstrated a clear trade-off between LIME's computational cost and explanation quality.

- Runtime: Increasing perturbation samples raised computational costs across all three models.
- Stability: Random Forest achieved the highest stability, reaching approximately 0.9.
- Fidelity: Random Forest also achieved the highest local fidelity (approximately 0.44 weighted R^2), but was the slowest model.

Limitations:

- Only one dataset (Covertype) was tested, with a fixed 54 features.
- Increasing samples did not consistently improve explanation quality.

Final Takeaway:
More samples do not necessarily produce better explanations. Choosing the right LIME configuration requires balancing runtime, stability and fidelity.

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